

Hammad Ahmad

AI / Machine Learning Engineer | LLMs · RAG · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

TECHNICAL SKILLS

Machine Learning, NLP & LLMs : PyTorch, scikit-learn, XGBoost, LightGBM, Retrieval-Augmented Generation (RAG), Sentence Transformers, vector embeddings, semantic search, cross-encoder re-ranking, RAGAS evaluation, Ollama

Engineering & Data : Python, TypeScript, JavaScript, SQL, FastAPI, Flask, REST APIs, React, Next.js, PostgreSQL, Neo4j, pandas, NumPy

MLOps & DevOps : MLflow, Docker, Git, GitHub Actions (CI/CD), Linux, AWS

EXPERIENCE

Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer (fixed-term contract)

Leeds, UK (Remote)

- Built and deployed an AI voice-agent system for outbound calling (Retell AI, FastAPI): backend dialogue-flow logic, automated call flows and lead-qualification rules, handling 2,100+ calls and cutting manual calling workload by roughly 50%.
- Reduced average call latency 54%, from 2.4s to 1.1s, by profiling backend async I/O and restructuring connection pooling; built an internal micro-CRM for lead tracking that removed external CRM licensing costs.

University of Bradford

Jan 2025 - Sep 2025

Research Assistant | Graph-Augmented LLM Engineering

Bradford, UK

- Built **FinLaw-UK** (code + benchmark), a graph-augmented RAG system for UK financial regulation: hybrid BM25 and BGE-small dense retrieval fused by reciprocal rank fusion, cross-encoder re-ranking, 2-hop Neo4j traversal, and a verifier resolving every citation against the graph before it reaches the user. Mistral 7B via Ollama, fully local.
- Re-measured the submitted evaluation and found the reported source-accuracy and citation-quality figures were regex shape-checks, not correctness measures: the scorer awarded a flat 0.85 to any citation-shaped string, and 103 of 110 rows carried that constant. Published the correction and the true graph-verified rate of 3 in 110, which motivated the citation normaliser, re-ranker and refusal gate.

COMSATS University Islamabad

Jul 2023 - Jul 2024

Research Intern, Data Science

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk on a 253,680-record CDC BRFSS dataset (14% positive class), comparing random over-sampling, SMOTE and ADASYN with resampling confined to training folds; Random Forest performed best on ROC-AUC and sensitivity. Deployed it as a public risk-screening web app (React, Flask): [DiabetesSense](#).

Quality Control Assistant, Myton Food Group (Morrisons), Bradford, UK

Apr 2026 - Present

PROJECTS

Jobzyl | [Live Platform](#) | Data Pipelines & Semantic Search · Python, FastAPI, Postgres

Jun 2026 - Present

- Own the retrieval stack behind a live job-search product: 20 upstream APIs ingested continuously into a **2M+ posting** Postgres index with cross-source deduplication, served through FastAPI over a semantic embedding index.
- Built the ATS-style CV matcher end to end: in-browser parsing so CVs are never uploaded whole, Fernet encryption at rest, and vector similarity with ranked skill-gap extraction for role-fit scoring.

VoiceFlow | [GitHub](#) | Speech Transcription Pipeline · FastAPI, PyTorch, Whisper

2026

- Built a FastAPI service running Whisper large-v3 inference in-process over **2,100+ production recordings**, dispatching each export as an OS thread to keep blocking downloads and torch inference off the asyncio event loop.
- Streamed live logs to the UI over Server-Sent Events alongside a polled progress endpoint, with CUDA/fp16 to CPU/fp32 device fallback, cooperative cancellation and a processed-ID manifest making re-runs idempotent.

EDUCATION

University of Bradford

Sep 2024 - Sep 2025

MSc, Artificial Intelligence (Merit)

Bradford, UK

- Dissertation**: FinLaw-UK, a graph-augmented retrieval system for UK financial regulation.

COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: [10.1007/978-3-031-66854-8_1](#)